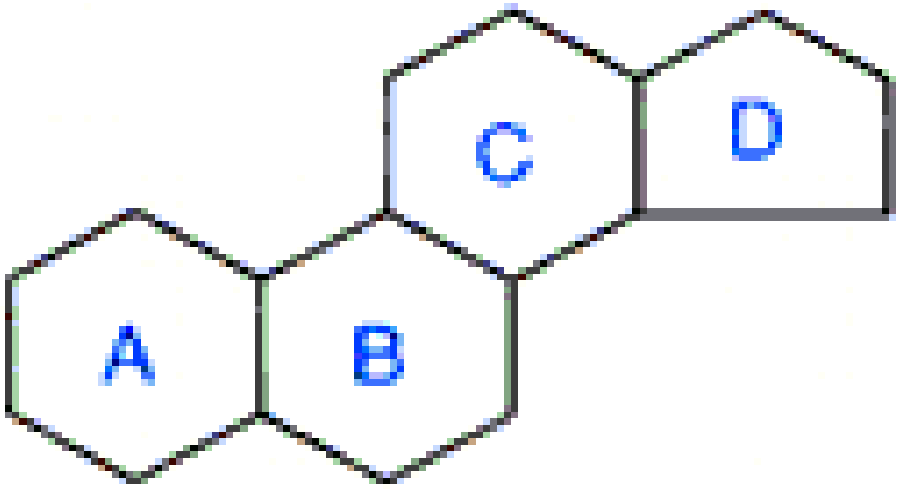
STEROIDS

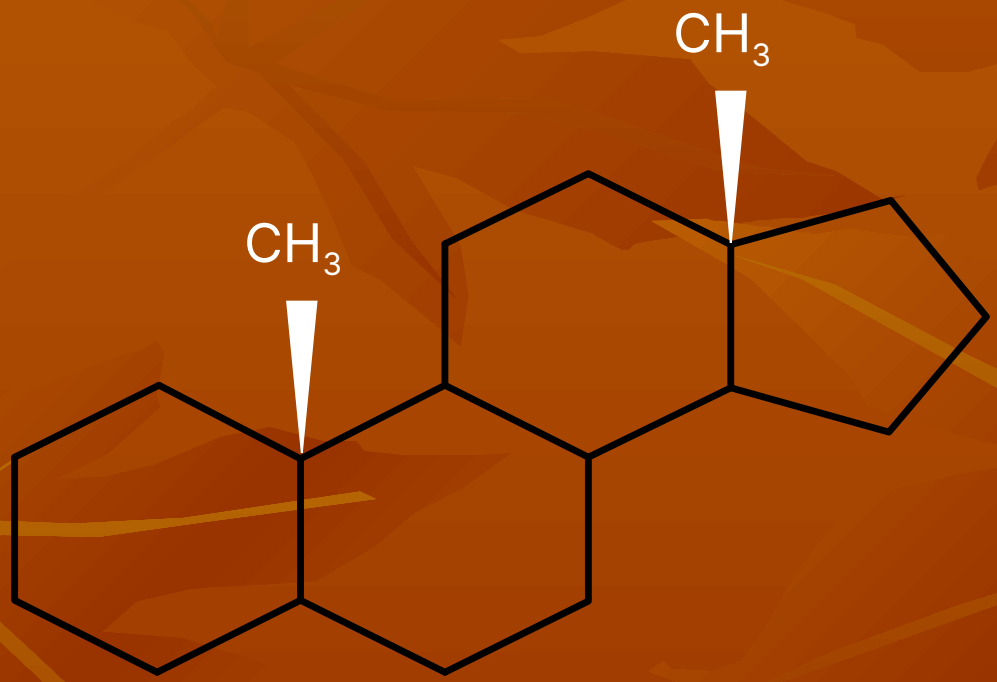
Dr. Muhammad Ahsan

STEROIDS

- **Compounds having a common characteristic structure based on the cyclopentanoperhydrophenanthrene**
- **3 cyclohexane rings**
- **1 cyclopentane ring**
- Total of 17 carbon atoms



Perhydrocyclopentanophenanthrene


CH3
CH3

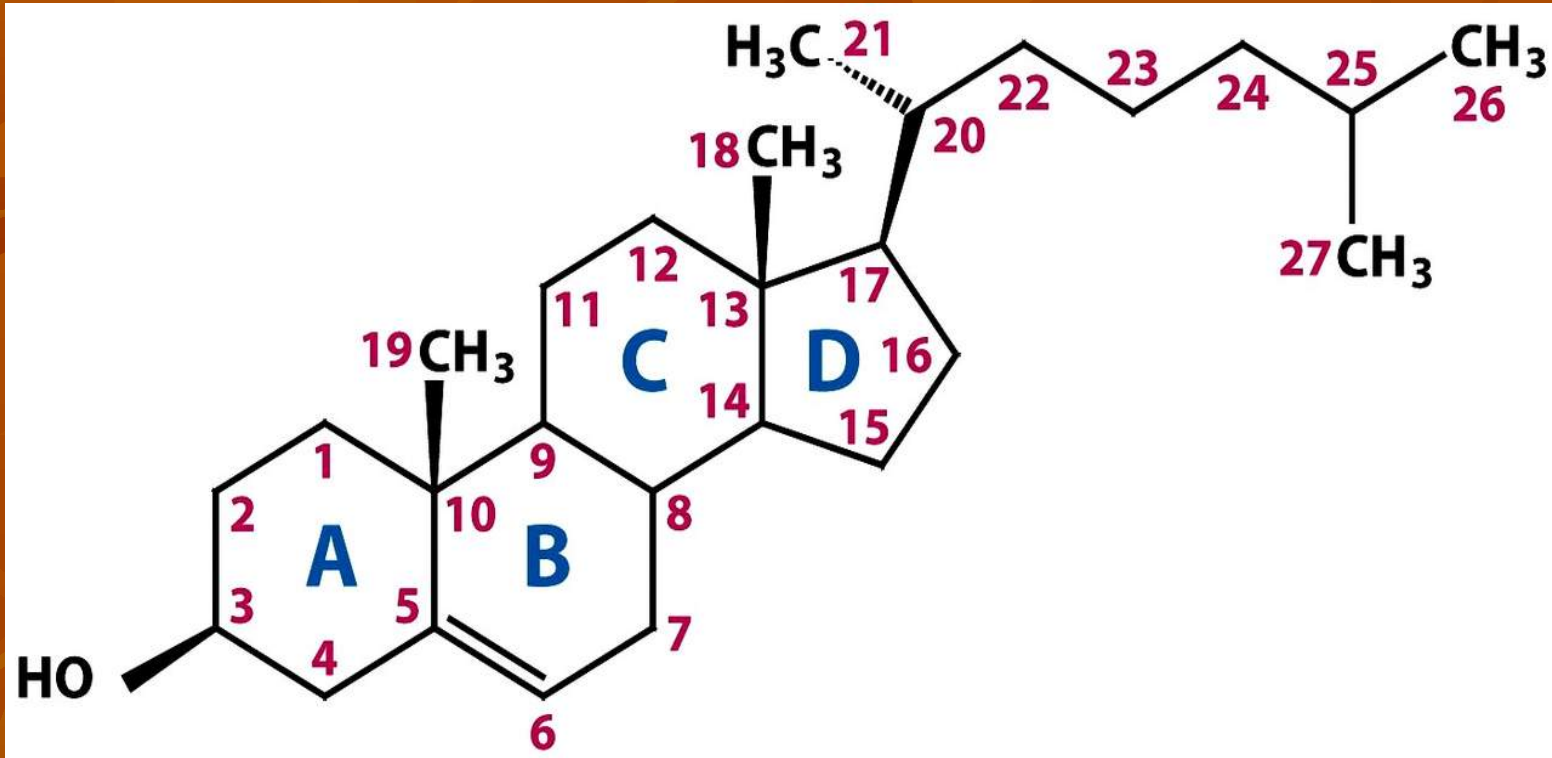


Figure 26-24
Biochemistry, Sixth Edition
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- Steroids are hydrocarbons with a multiple ring structure
- They are hydrophobic and lipophilic
- Contain no fatty acids
- Cholesterol is the best-known steroid, found only in animal products

- Various substances of physiological interest having the same steroid structure are:

- STEROLES

  Cholesterol

  7-Dehydrocholesterol

  Ergosterol

  Calciferol

  Coprosterol

- BILE ACID
- SEX HORMONES

Female: Estrogen
Progesterone
Male: Testosterone
Androsterone

- ADRENOCORTICAL HORMONES
- SAPONINS
- CARDIAC GLUCOSIDES
- TOAD POISON

Functional Classification of Steroids

- **Anabolic Steroids**
  - Interact with androgen receptor; enhance muscle mass/athlete's performance; male sex hormones
- **Glucocorticoids**
  - regulate metabolism and immune function; anti-inflammatory activity
- **Mineralocorticoids**
  - maintain blood volume and renal excretion
- **Progestins**
  - Development of female sex organs and characteristics
- **Phytosteroids**
  - Plant steroids
- **Ergosteroids**
  - Steroids of the fungi; vitamin D related

STEROLE

- Sterol means solid alcohol
- Steroids with 8-10 carbon atoms in the side chain at position 17 and an alcoholic hydroxil group at position 3
- Widely present in animal and plant tissues
- Among the sterols Cholesterol forms an important physiological compound

Cholesterol

- **Cholesterol means solid bile alcohol**
- **Derived from the Greek word ‘Cholesterine’ (Chole-bile, sterine-solid)**
- **Most abundant steroid in the body**
- **Cholesterol occurs in animal fats and not in plants**
- **First isolated from human gall stones**

Source of Cholesterol

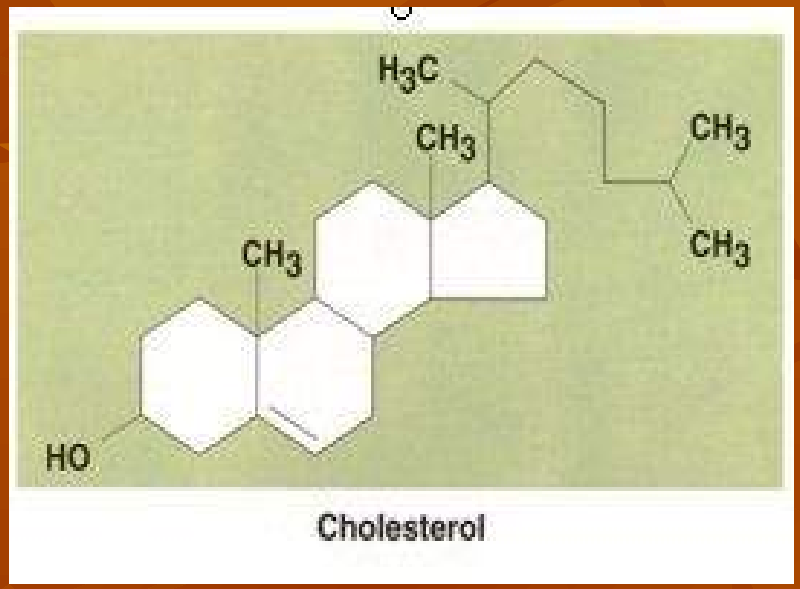
- **Myelin sheath, brain, and nerve tissue**
- **Adrenal glands, Glandular tissues**
- **Plasma**
- **Cellular membranes**
- **Egg yolk**



Structure of Cholesterol

- Composed of the characteristic cyclopentanoperhydrophenanthrene ring system

- In addition to the ring system cholesterol has long aliphatic chain attached to C_{17} and an alcoholic OH group attached to C_3

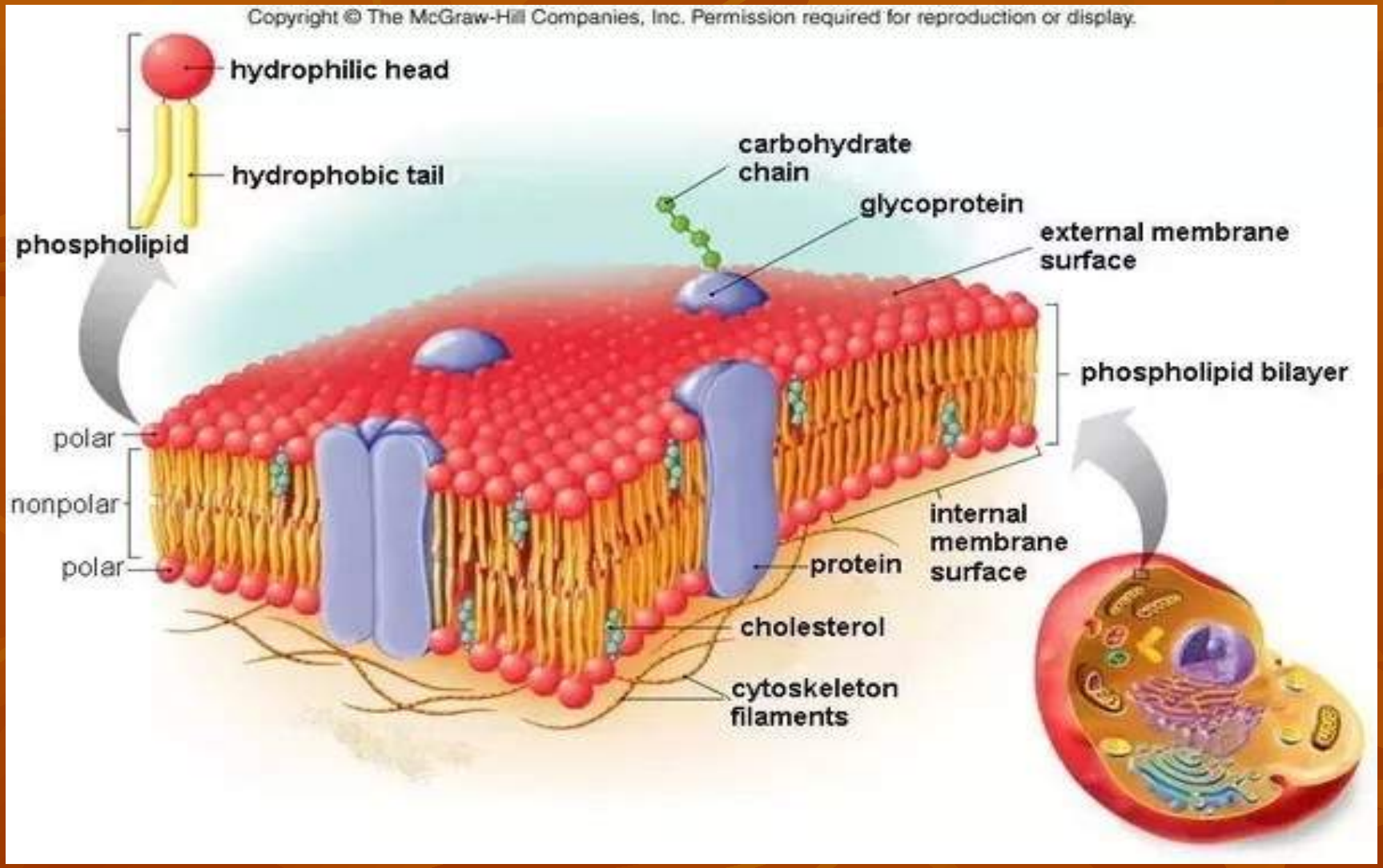


Cholesterol

- Structure reveals 8 asymmetric carbon atoms in positions 3,5,8,9,10,13,14 and 17
- This means there are 2^8=256 sterioisomers of cholesterol
- Molecular formula of cholesterol is $C_{27}H_{45}OH$
- It is a monounsaturated alcohol because of the presence of a double bond at C_5-C_6

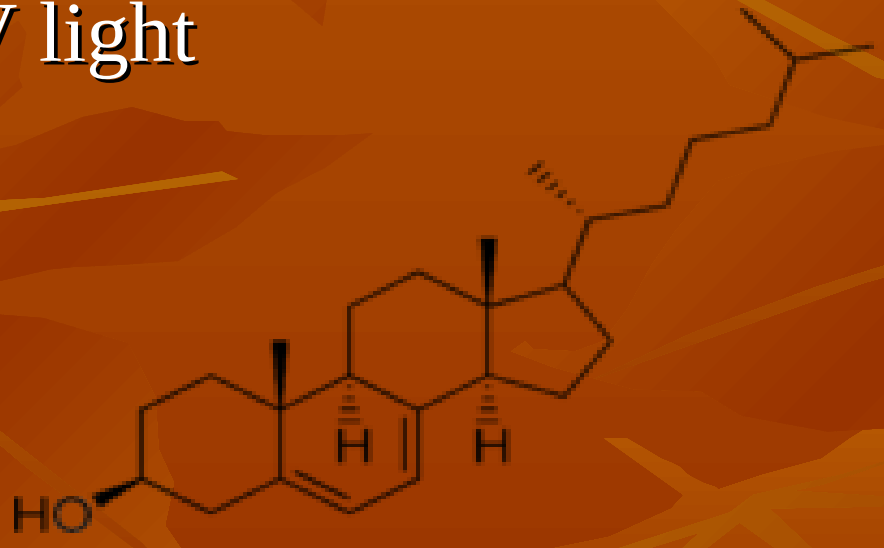
Cholesterol: Functions

- Major component of cell membranes (especially abundant in nerve and brain tissue)
- Precursor molecule: Example - Vitamin D and estrogen are synthesized from cholesterol
- Important in the synthesis of bile acids


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hydrophilic head
hydrophobic tail
phospholipid
carbohydrate chain
glycoprotein
external membrane surface
phospholipid bilayer
polar
nonpolar
polar
internal membrane surface
protein
cholesterol
cytoskeleton filaments

7-Dehydrocholesterol

- Important sterol present in skin
- It has one more double bond at C_7-C_8
- It is converted into vitamin D when irradiated with UV light



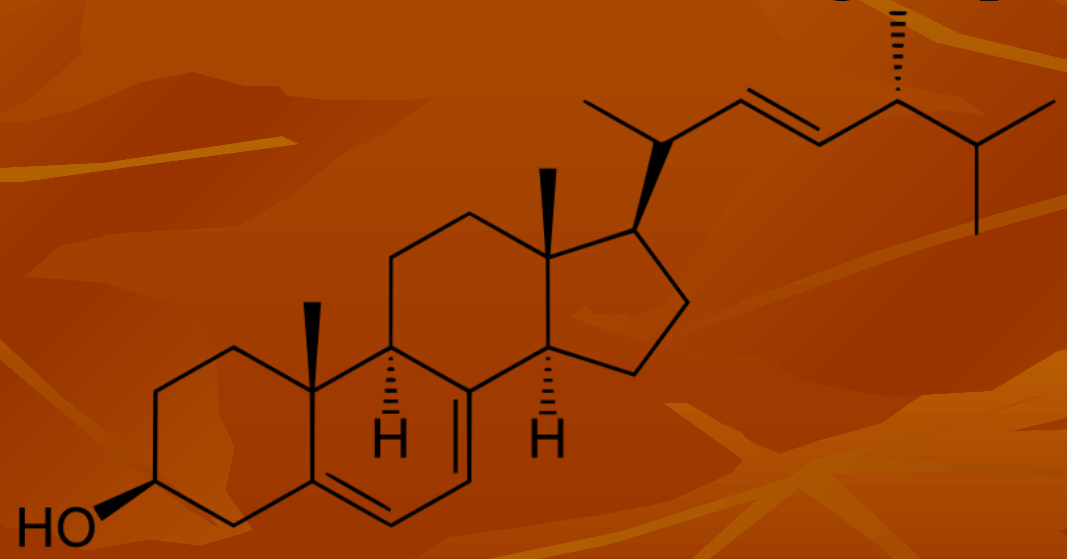
Ergosterol

- Found in fungi
- First isolated from ergots (common name for the members of the fungal genus *Claviceps*)
- Dose not occur in plant or animal cells
- Component of yeast and fungal cell membranes
- Biological precursor to vitamin D2,can be converted by irradiation with UV light

- Structure is similar to that of 7-dehydrocholesterol,but differs in the following aspects:

  i) it has a double bond in the side chain

  ii) it has an additional CH_3 group in the side chain

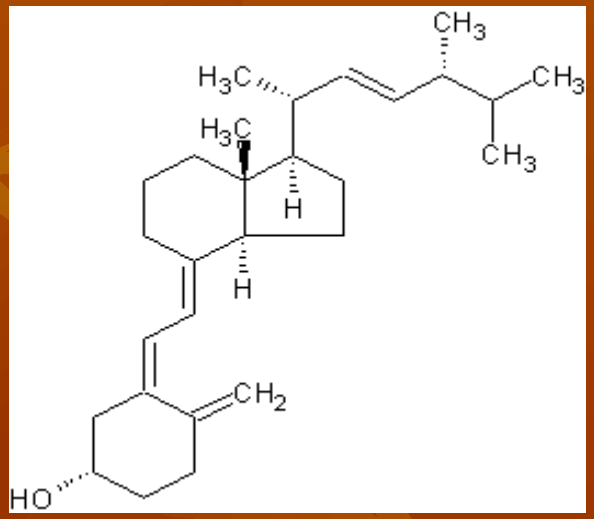


Vitamin D (Calciferol)

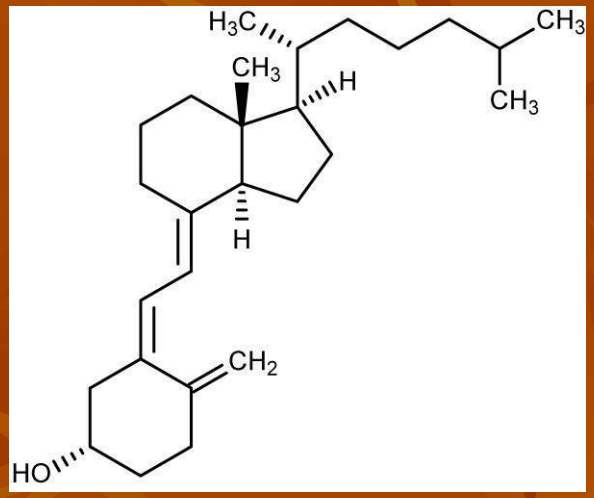
- Chemically vit D are secosteroids i.e.,steroids in which one of the bonds in the steroid ring is broken
- Body can synthesis it when sun exposure is adequate (hence its nick name the “sunshine vitamin”)
- Several forms exist
- Two major forms are vit D_2 or ergocalciferol (made from ergosterol) and vit D_3 or cholecalciferol (made from 7-dehydrocholesterol)

- The structural difference between vit D_2 and vit D_3 is in their side chains
- Side chain of D_2 contain a double bond between C_{22} and C_{23}, and a methyl group on C_{24}

VITAMINE D2



VITAMINE D3



Vitamin D: Functions

- It promotes absorption and use of calcium and phosphate for healthy bones and teeth
- It prevents rickets in children and osteomalacia in adults, helps to protect older adults from osteoporosis
- Also affects neuromuscular function, and influences action of many genes that regulate proliferation, differentiation and apoptosis of cells

Coprosterol

- Present in faeces
- Reduced form of cholesterol
- Formed as a result of reduction by intestinal bacteria
- It does not have the double bond between C5 and C6

BILE ACIDS

- Synthesized in liver from cholesterol
- Stored in the gall bladder
- Contain additional carboxyl group in the side chain
- Four bile acids have been isolated from human bile

i) Cholic acid

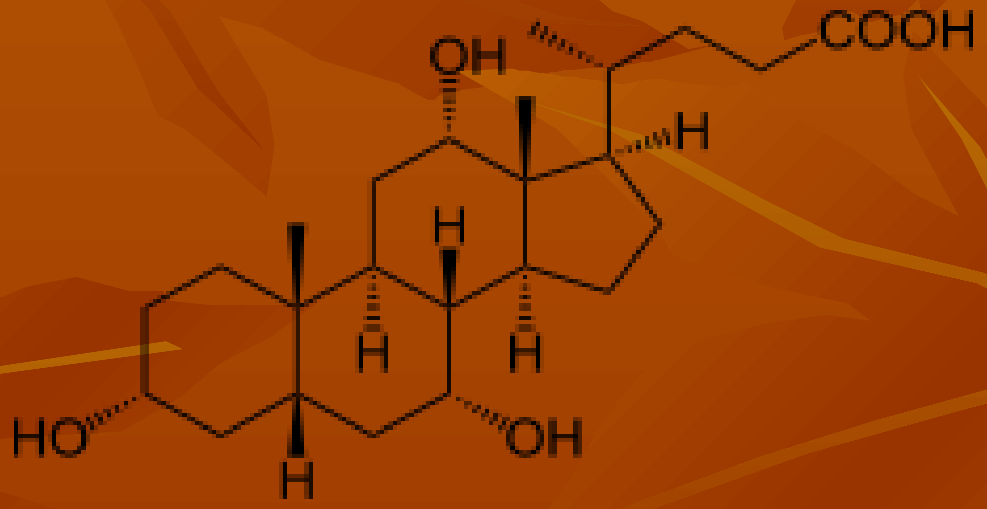
ii) Chenodeoxycholic acid

iii) Deoxycholic acid

iv) Lithocholic acid



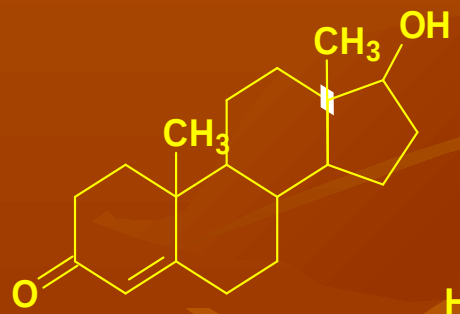
- Cholic acid



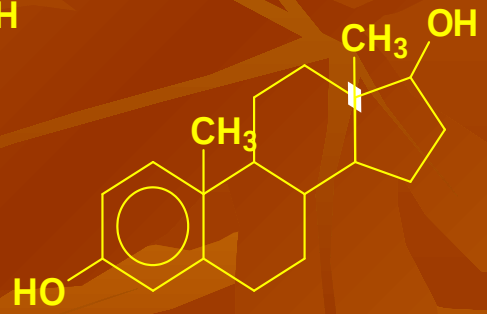
SEX HORMONES

- **Androgens (testosterone and androsterone) in males**
- **Estrogens (estron,estradiol and estriol) and Progesterone in females**





testosterone



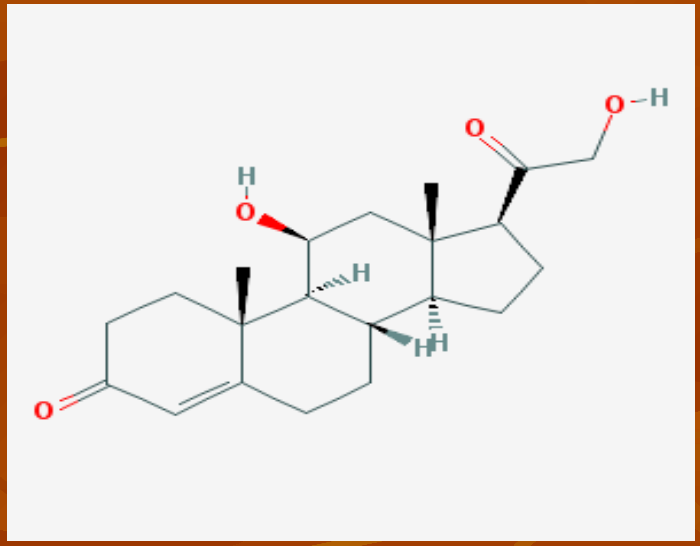
estradiol

- They do not have side chain in position 17
- They possess either a keto or hydroxyl group in positions 3 and 17
- Estron has a keton group in position 17
- Estradiol has a hydroxyl group in position 17
- Estriol has hydroxyl group in position 16 and 17
- Progestrone has ketone groups at C3 and C17

ADRENOCORTICO HORMONES

- Corticosteron contains ketone group in positions C_3 and C_{17}
- It has a OH group at C_{11}

CORTICOSTERONE



- SAPONINS are hemolytic poisons widely distributed in plants
- CARDIAC GLUCOSIDES are present in digitalis and squill
- TOAD POISON have the structure of steroids. They are cardiac toxins secreted by the parotid glands of toads

Thank you…